Probabilistic Boolean circuit axioms are complete for Boolean stochastic maps

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Editorial note. Copyright Alessandro Di Giorgio, Pawel Sobocinski and Niels Voorneveld; official CSL2026.29 CC BY4.0. Main Theorem11 and its AppendixB Theorem23 restatement count once. All original probabilistic Boolean circuit diagrams and exact printed axiom tags retained; normal-form support and metric applications do not add counts.

Allowed standard prerequisites (editorial).

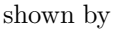
- Symmetric monoidal/Markov-category coherence, finite probability distributions and stochastic maps. Original Section2 and bibliography, with full current probabilistic-choice axioms and all current normal-form constructions retained. No metric/enrichment or external new-core proof is needed for selected completeness target.

2 Symmetric Monoidal Categories as Resource Theories

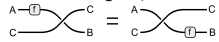
In this section we give a brief introduction to symmetric monoidal categories (SMCs) and their associated graphical calculus [13, 23, 22] in terms of resource theories, following [6].

The basic data consists of objects $A, B, C, \ldots$ that we think of as types of resources, and morphisms $f\colon A \to B$, which we think of as transformations that convert a resource of type A into one of type B. Morphisms can be composed sequentially: given $f\colon A \to B$ and $g\colon B \to C$, their composite $f;g\colon A \to C$ represents performing f followed by g.

The monoidal product $\otimes$ lets us express having two resources at once – so $A \otimes C$ means having both A and C – and it extends to morphisms too: given $f\colon A \to B$ and $g\colon C \to D$, the morphism $f \otimes g\colon A \otimes C \to B \otimes D$ describes applying f and g in parallel. There is also a void resource I that acts neutrally under combination: $A \otimes I = A$.

These ideas are captured visually using the graphical calculus of string diagrams. A resource type A is drawn as a wire , and a transformation $f\colon A_1 \otimes \ldots \otimes A_n \to B_1 \otimes \ldots \otimes B_m$ is represented by a box  with n input wires on the left and m output wires on the right. Composition of morphisms is shown by connecting boxes in sequence  , while the monoidal product corresponds to placing them side-by-side  . The void resource I corresponds to no wires at all – i.e., the empty diagram.

Resources can also be swapped according to the symmetry $\chi_{A,B}\colon A \otimes B \to B \otimes A$, which is represented by a crossing of wires  .

Diagrams are subject to the axioms of symmetric monoidal categories (SMCs). However, note that equational reasoning in this setting amounts to deformation of diagrams without changing their topology. For example, naturality of symmetry amounts to sliding a box past a crossing of wires, illustrated as  .

In the next section, where we introduce the parametric iteration construction, it will be convenient to explicitly track both the combination of resources via the monoidal product and the void resource. Therefore, following [4], we extend the graphical calculus with operations that introduce  and discard the void resource, as well as split  and combine  resources. These operations are subject to the expected equations determined by the properties of the monoidal product.

Finally, note that all monoidal categories considered in this paper are taken to be strict, as justified by Mac Lane's strictification theorem [18].

4 Reasoning with Probabilities and Distances

To model richer computational behaviors, we consider symmetric monoidal categories with additional properties and structure. In particular, for probabilistic computations, we focus on metric-enriched symmetric monoidal categories and Markov categories [10].

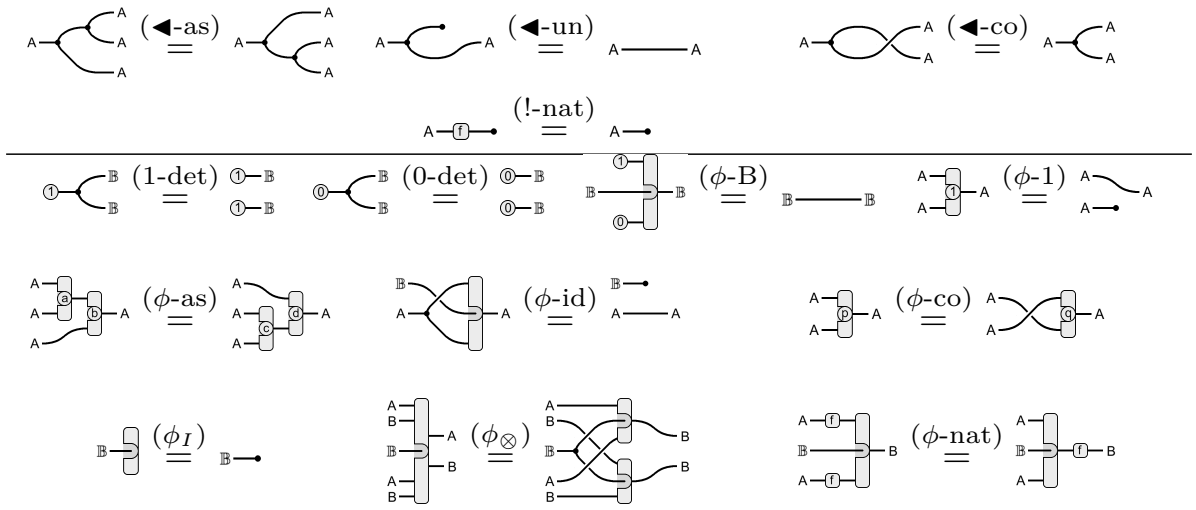
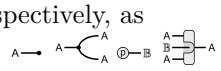


Figure 1: Axioms for Markov categories (top) and probabilistic choice (bottom). Here $f\colon A \to B$, $q = 1 - p$, $c = \frac{(1-a)b}{1-ab}$ and $d = ab$.

Definition 9.

A Markov category with probabilistic choice is a symmetric monoidal category equipped with operations $!_A\colon A \to I$, $◀_A\colon A \to A \otimes A$, $\langle p\rangle\colon I \to \mathbb{B}$ and $\phi_A\colon A \otimes \mathbb{B} \otimes A \to A$, represented in string diagrams, respectively, as



where $p \in [0, 1]$, $\mathbb{B}$ is a designated object and A is any object, and such that the axioms in Figure 1 hold. That is:

■ $(◀_A, !_A)$ is a cocommutative comonoid, that is $◀_A$ is associative ($◀$ -as), cocommutative ($◀$ -co) and $!_A$ is the unit ($◀$ -un),

■ $◀_A$ and $!_A$ are coherent with the monoidal structure,

■ $!_A$ is natural (!-nat),

■ $\langle 1\rangle$ and $\langle 0\rangle$ are deterministic ((1-det) and (0-det)),

■ ϕ_A is coherent with the monoidal structure ((ϕ_I) and ($\phi_\otimes$)),

■ ϕ_A is natural in A (ϕ-nat),

■ ϕ_A^p–where ϕ_A^p is  , often simply  –is a convex algebra, that is, ϕ_A^p is idempotent (ϕ-id) and satisfies parametric associativity (ϕ-as) and commutativity (ϕ-co),

■ ϕ and $\langle 1\rangle$, $\langle 0\rangle$ interact via (ϕ-B) and (ϕ-1).

The $\mathbb{B}$ represents the type of two bit values, 1 and 0. $\langle p\rangle$ describes a weighted coin flip, which has a chance of p of producing 1, and a chance of $1 - p$ of producing 0. The ϕ represents an if-gate, choosing between two results of the same type using a bit as an argument.

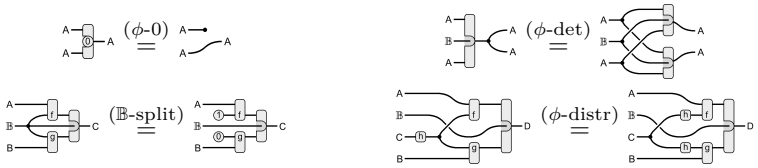


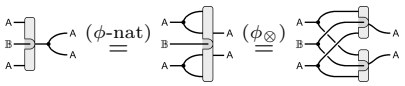
Figure 2: Derived laws for Markov categories with probabilistic choice.

Lemma 10.

The laws in Figure 2 hold in any Markov category with probabilistic choice.

Proof.

We give a proof only for (ϕ-det), the rest can be found in Appendix C.



◀ Our leading example of Markov category with probabilistic choice is $\mathsf{FinStoch}$, the category of sets and stochastic maps. Objects are sets and morphisms $f\colon A \to B$ are functions $A \to \mathcal{D}B$ mapping each $a \in A$ to a finitely-supported discrete probability distribution over B. We represent a distribution as a function $v : A \to [0,1]$ with finite support $\mathtt{supp}(v) := \{a \in A \mid v(a) > 0\}$ and such that $\sum_{a\in A} v(a) = 1$, or, alternatively, as a formal sum $\sum_i p_i \,|a_i\rangle$, where the $|\cdot\rangle$ notation is used to separate the elements $a_i \in A$ from their associated probability $p_i \in [0,1]$.

We write $f(b|a)$ for $f(a)(b)$, representing the probability that f returns b, given a. For two stochastic maps $f\colon A \to B, g\colon B \to C$, their composition is defined by summing over the middle variable, i.e. $(f;g)(c \mid a) := \sum_{b\in B} g(c \mid b)f(b \mid a)$. Identities $id_A\colon A \to A$ map each $a \in A$ to the Dirac distribution at a, i.e. $id_A(a) := |a\rangle$.

Given two distributions $v \in \mathcal{D}A$ and $u \in \mathcal{D}B$, their product $v\otimes w \in \mathcal{D}(A \times B)$ is given by $v\otimes u(a,b) := v(a)u(b)$. This yields a symmetric monoidal structure, with the monoidal product $\otimes$ being the cartesian product of sets $\times$ on objects and with monoidal unit the singleton set $I := \{\bullet\}$. For arrows $f\colon A \to B$ and $g\colon C \to D$, $f \otimes g\colon A \otimes C \to B \otimes D$ is defined as $f \otimes g(b,d|a,c) := f(b|a)g(d|c)$, and symmetries $\chi_{A,B}\colon A \otimes B \to B \otimes A$ as $\chi_{A,B}(a,b) := |b\rangle \otimes |a\rangle$.

The comonoid structure is given by $◀_A\ (a) := |a\rangle \otimes |a\rangle$ and $!_A(a) := |\bullet\rangle$, while for the probabilistic choice structure we take as choice object the set of Booleans $\{1,0\}$ and, for any $p \in [0,1]$, we define $\langle p\rangle\colon \{\bullet\} \to \{1,0\}$ as $\langle p\rangle(\bullet) := p\,|1\rangle + (1-p)\,|0\rangle$ and $\phi_A\colon A \otimes \{1,0\} \otimes A \to A$ as $\phi_A(a,1,b) := |a\rangle$ and $\phi_A(a,0,b) := |b\rangle$. Note that $\phi^p_A(a,b) = p\,|a\rangle + (1-p)\,|b\rangle$.

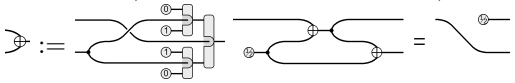
Throughout the rest of the paper we will work in $\mathsf{FinStoch}_{\{1,0\}}$, the subcategory of $\mathsf{FinStoch}$ consisting of stochastic maps between sets that are powers of the booleans. The structure of Markov category with probabilistic choice, introduced here, is novel and it offers an alternative presentation of $\mathsf{FinStoch}_{\{1,0\}}$ to the one in [21]. In particular:

Theorem 11.

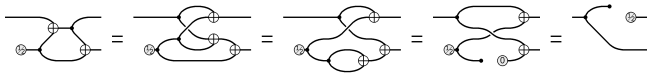
the axioms in Figure 1 are complete for $\mathsf{FinStoch}_{\{1,0\}}$.

The interested reader can find the details in Appendix B.

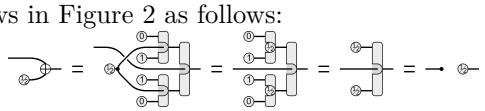
As an example of diagrammatic reasoning in $\mathsf{FinStoch}_{\{1,0\}}$ we look at the one-time-pad protocol on a single bit. We regard $\langle\frac{1}{2}\rangle\colon I \to \mathbb{B}$ as an operation that produces a randomly generated bit, and the xor $\oplus\colon \mathbb{B} \otimes \mathbb{B} \to \mathbb{B}$, defined below on the left, as an operation that encrypts and decrypts a bit.



The equation on the right expresses the correctness of the protocol: encoding and decoding with the same randomly generated bit gets the same result, and that to an adversary the encrypted message looks completely random. We prove correcteness by equational reasoning:



The equations use the fact that $\oplus$ is deterministic, associative, idempotent, unitary with respect to $\langle 0\rangle$ and nullary with respect to $\langle\frac{1}{2}\rangle$. All of these properties can be proved via the axioms of Markov categories with probabilistic choice. For example the fact that $\langle\frac{1}{2}\rangle$ is absorbing for $\oplus$ is proved using the laws in Figure 2 as follows:



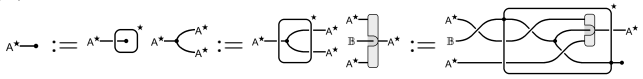
Looking at $\mathsf{PI}(\mathsf{FinStoch}_{\{1,0\}})$, the above example can be extended to operations on $\mathbb{B}^\star$, encrypting and decrypting $\star$-tuples of bits. We now show that the parametric iteration construction preserves the structure of Markov categories with probabilistic choice.

Lemma 12.

If $\mathcal{C}$ is a Markov category with probabilistic choice, then so is $\mathsf{PI}(\mathcal{C})$.

Proof.

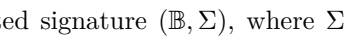
For $\mathbb{B}$ we can take $(\!|\mathbb{B}|\!)$, and for $\langle p \rangle$ we take $(\!|\langle p \rangle|\!)$. We define discard $!_{A^\star}$, copy $\blacktriangleleft_{A^\star}$ and case $\phi_{A^\star}$ for $\mathsf{PI}(\mathcal{C})$ inductively on objects, with their definition on $A^\star$ being:



The relevant properties can be shown using the diagonal method, with the naturality equations in particular being instances of Newton's cradle. ◀

Appendix B A Complete Equational Theory of Boolean Stochastic Maps

The structure of Markov categories with probabilistic choice, introduced in Section 4, is novel. Thus, it is interesting to study whether the axioms are complete with respect to the category $\mathsf{FinStoch}_{\{1,0\}}$. Here, completeness means that any equality between Boolean stochastic maps in $\mathsf{FinStoch}_{\{1,0\}}$ can also be derived syntactically using the axioms listed in Figure 1.

In order to formally state completeness, we introduce the syntactic category PBCirc of probabilistic boolean circuits. This is the symmetric monoidal category of string diagrams freely generated by the single-sorted signature $(\mathbb{B}, \Sigma)$, where $\Sigma := \{$  $\}$ and such that the morphisms are quotiented by the axioms in Figure 1.

Diagrams in PBCirc take semantics in $\mathsf{FinStoch}_{\{1,0\}}$ via the interpretation functor $[\![-]\!]\colon \mathsf{PBCirc} \to \mathsf{FinStoch}_{\{1,0\}}$, defined inductively as follows:

$$[\![\mathbb{B}]\!] := \{1,0\} \quad [\![\,\mathbb{B}\!\rightarrow]\!] :=!_{\{1,0\}} \quad [\![\,\mathbb{B}\!\leftarrow\!\mathbb{B}\,]\!] :=\blacktriangleleft_{\{1,0\}} \quad [\![\,\mathbb{B}\!\leftarrow\!\mathbb{B}\,]\!] := \phi_{\{1,0\}} \quad [\![\,\mathbb{O}\!\leftarrow\!\mathbb{B}\,]\!] := \langle p \rangle$$

Then we say that the equational theory of PBCirc is complete for $\mathsf{FinStoch}_{\{1,0\}}$ if, for any two diagrams $c, d \in \mathsf{PBCirc}$ of the same type, if $[\![c]\!] = [\![d]\!]$ then their equality can be derived using the axioms in Figure 1. We rephrase Theorem 11.

Theorem 23.

The equational theory of PBCirc is complete for $\mathsf{FinStoch}_{\{1,0\}}$.

Proof.

The proof shows that each element of $\mathsf{FinStoch}_{\{1,0\}}$ has a unique normal form representation as a diagram in PBCirc, and each diagram in PBCirc is provably equal to a diagram in normal form.

We first equip each set $\mathbb{B}^n$ with a linear order. We then say that a formal sum $p_1 \cdot x_1 + \cdots + p_m \cdot x_m$ over $\mathbb{B}^n$ is in normal form if all $p_i > 0$ and the list $x_1, \ldots, x_m$ forms a sorted list of elements without repetition. As such, a formal sum in normal form is either of the form $1 \cdot x$, or $p \cdot x + (1-p) \cdot s$ with $p \in (0,1)$, and s a formal sum in normal form, containing values x' all higher than x according to the chosen order. Importantly, each distribution has a unique normal formal sum representation.

A weighted probabilistic tree over $\mathbb{B}^n$ is a diagram $I \to \mathbb{B}^n$ inductively defined as:

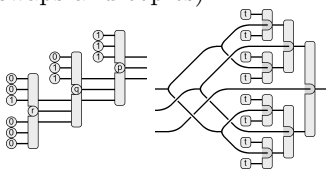
$$t, r \quad := \quad \langle b_1 \rangle \otimes \cdots \otimes \langle b_n \rangle \mid (t \otimes r); \phi^p_{\mathbb{B}^n}$$

where each $b_i \in \{0,1\}$. We represent a normal formal sum as a weighted probabilistic tree, by representing $1 \cdot (b_1, \ldots, b_n)$ as $\langle b_1 \rangle \otimes \cdots \otimes \langle b_n \rangle$, and $p \cdot (b_1, \ldots, b_n) + (1-p) \cdot s$ as $(\langle b_1 \rangle \otimes \cdots \otimes \langle b_n \rangle \otimes t); \phi^p_{\mathbb{B}^n}$, where t is the representation of the normal formal sum s. It can be shown that any weighted probabilistic tree over $\mathbb{B}^n$ is equal to the tree representation of a normal formal sum. This is done by using axioms (ϕ-1), (ϕ-co) and (ϕ-as) to get rid of zero-probability branches, sorting the order, and re-associating branches, as well as determinism of $\langle 1 \rangle$, $\langle 0 \rangle$ combined with (ϕ-id) to combine branches with the same result.

The normal forms of type $\mathbb{B}^0 \to \mathbb{B}^n$ are the tree representations of normal formal sums, which are unique for each distribution. We generalize to arbitrary $\mathbb{B}^k \to \mathbb{B}^n$ by doing a case distinction on inputs, where a normal form of type $\mathbb{B}^{k+1} \to \mathbb{B}^n$ is of the form: $(\blacktriangleleft_{\mathbb{B}^k} \otimes id_{\mathbb{B}}); (id_{\mathbb{B}^k} \otimes \chi_{\mathbb{B}^k,\mathbb{B}}); (f_1 \otimes id_{\mathbb{B}} \otimes f_0); \phi_{\mathbb{B}^n}$, where f_1 and f_0 are normal forms of morphisms of type $\mathbb{B}^k \to \mathbb{B}^n$. Each morphism $f : \mathbb{B}^k \multimap \mathbb{B}^n$ of $\mathsf{FinStoch}_{\{1,0\}}$ has a unique normal form of this kind, with $f : \mathbb{B}^0 \to \mathbb{B}^n$ represented by the tree representation of the normal formal sum for the distribution given by $f(*)$, and $f : \mathbb{B}^{k+1} \to \mathbb{B}^n$ represented by taking f_1 and f_0 to be the normal form of the functions $(b_1, \ldots, b_k) \mapsto f(b_1, \ldots, b_k, 1)$ and

$(b_1, \ldots, b_k) \mapsto f(b_1, \ldots, b_k, 0)$ respectively. This normal form is unique, as this specification is necessitated by the $[\![-]\!]$ functor.

See below examples of normal forms, with a normal form of a morphism of type $\mathbb{B}^0 \to \mathbb{B}^3$ on the left in the form of a sorted weighted tree (using lexicographic ordering on $\mathbb{B}^3$), and a normal form of a morphism of type $\mathbb{B}^3 \to \mathbb{B}^k$ with a weighted tree at each t (some liberties were taken regarding the order of swaps and copies).



We prove that each diagram is equivalent to one in normal form.

■

First we show that the identity diagrams $id_{\mathbb{B}^n} : \mathbb{B}^n \to \mathbb{B}^n$ can be put in normal form. This can be shown by using the equations (ϕ-B) and (ϕ-id) together with monoidal coherence and naturality of ϕ, given in equations (ϕ_I), ($\phi_\otimes$), and (ϕ-nat).

■

Second, note that each morphism in a freely generated monoidal category can be described as a composition of morphisms of the form $(id_A \otimes s \otimes id_B)$, with s a generator, which in our case is of $\{\chi\} \cup \Sigma = \{\chi, \blacktriangleleft, !, \langle p \rangle, \phi\}$. Then show that $f; (id_A \otimes s \otimes id_B)$ can be reduced to a normal form, given that f is in normal form. This is done in three steps:

1.

Pull $(id_A \otimes s \otimes id_B)$ through each ϕ using naturality (ϕ-nat), both the if-gates used to distinguish between different inputs, as well as the if-gates used in the weighted probabilistic trees, until we get to leaves of the trees.

2.

Reduce each $(\langle b_1 \rangle \otimes \cdots \otimes \langle b_n \rangle); (id_A \otimes s \otimes id_B)$ to a weighted probabilistic tree.

3.

The result is f, but with different weighted probabilistic trees at each case in its case distinction. Put all those trees into normal form.

Hence we can normalize each $f : \mathbb{B}^k \to \mathbb{B}^n$ by first giving it a specification as composition of $(id_A \otimes s \otimes id_B)$ morphisms. Then normalize the identity function on $\mathbb{B}^k$ and inductively normalize by composing with each additional layer. ◀

Lemma 24.

PBCirc is a metric enriched symmetric monoidal category.

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